MILITARY SPECIFICATION

RUBBERS, SYNETHETIC, FOR CHEMICAL AGENT COMPOUNDING

1. SCOPE

1.1 This specification covers two types of synthetic rubbers employing styrene and butadiene polymers.

1.2 Classification. The synthetic rubber shall be furnished in the following types and classes as specified (see 6.2):

Type I - SBR type 1000

Class 1 - Baled.

Type II - SBR type 1006

Class 1 - Baled.
Class 2 - Crumbed.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following document of the issue in effect on date of invitation for bids or request for proposal forms a part of this specification to the extent specified herein:

STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM Standards

D1416-61T - Chemical Analysis of Synthetic Elastomers (Solid Styrene - Butadiene Copolymers.
D1419-61T - Recommended Practices for Description of Types of Styrene-Butadiene Rubbers (SBR) and Butadiene Rubber (BR).
D1646-61 - Test for Viscosity and Curing Characteristics of Rubber by the Shearing Disk Viscometer.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Ratings, Rules and Regulations.

(Application for copies of these freight classification rules should be addressed to the Uniform Classification Committee, 202 Union Station, Chicago 6, Illinois.)

3. REQUIREMENTS

3.1 Material. The synthetic rubber shall consist of styrene and butadiene polymers suitably compounded according to type as described in ASTM D1419-61T. Material shall be no older than 6 months.

3.2 Chemical and physical properties. The chemical and physical properties for type I rubber (SBR 1000) and type II (SBR 1006) shall be as specified in table I when tested as specified therein.

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Table I. Chemical and physical properties

Property	Requirement		Test method
	Type I	Type II	
Volatile matter, %	0.50 max	0.50 max	4.6
Ash, %	1.50 max	1.50 max*	4.6
Fatty acid (as stearic acid), %	4.00 to 6.25	3.75 to 6.00	4.6
Soap, %	0.75 max	0.75 max	4.6
Stabilizer, %	1.00 to 1.75	1.00 to 1.75	4.6
Bound styrene, %	22.5 to 24.5	22.5 to 24.5	4.6
Mooney viscosity, ML 212°F at 4 minutes	46 to 52	46 to 54	4.6
Viscosity of compounded stock, ML 212°F at 4 minutes	73 max	73 max	**
Tensile strength, psi, 50-minute cure at 292°F	2700 min	2500 min	**
Ultimate elongation, 50 minute cure at 292°F	550 min	500 min	**
Modulus, psi, 300% elongation			
25 minute cure at 292°F	300 to 600	400 to 700	**
50 minute cure at 292°F	800 to 1150	900 to 1250	**
100 minute cure at 292°F	1250 to 1650	1400 to 1800	**

* When type II, class 2 material is specified, specify the ash content if different (see 6.2).

** These properties shall be determined for information only using the standard commercial test methods.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Special provisions.

4.2.1 Alternative inspection. The supplier may utilize any alternative inspection procedure which will provide equal or better assurance of quality by submitting a written proposal with justification and obtaining written approval from the Government prior to instituting the procedure. In case of dispute, the procedures of this specification will govern.

4.2.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.3 Lotting. A lot shall consist of the rubber of one type and class with the same code indicating one day's production from a single drier.

4.4 Sampling. A random sample of at least 1,000 grams shall be taken from each of four bales or an equivalent quantity of crumbed rubber, in the lot. The samples shall be taken in such a manner that they are free from dusting material e.g., from the inside of the bales and immediately placed in clean, dry, air-tight containers.

4.5 Inspection provisions.

4.5.1 Volatile matter. Samples shall be tested individually as specified in 4.6, and if the average is greater than the specified limit or if any single value is in excess of twice the specified limit, the lot shall be rejected.

4.5.2 Properties other than volatile matter. Samples shall be tested individually as specified in 4.6, and if any individual sample fails to meet the specified limits, the lot shall be rejected.

4.6 Tests. Tests shall be conducted in accordance with table II using the appropriate ASTM test method.

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Table II. Test methods

Requirement	ASTM Method
Volatile matter	D1416-61T
Ash	D1416-61T
Fatty acid	D1416-61T
Soap	D1416-61T
Stabilizer	D1416-61T
Bound styrene	D1416-61T
Viscosity	D1646-61

5. PREPARATION FOR DELIVERY

5.1 Packaging, level C. The rubber of one type shall be packaged to afford adequate protection from the supply source to the first receiving activity for immediate use.

5.2 Packing, level C. Rubber of one type, packaged as specified in 5.1, shall be packed to insure carrier acceptance and safe delivery to the first domestic destination. Containers shall comply with Uniform Freight Classification rules or regulations of other carriers applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required by the contract or order, unit packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The synthetic rubbers covered by this specification are intended for use in compounding plasticized white phosphorus.

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type and class of rubber required.
(c) Ash content if different from table I.

6.3 Significant places. For the purpose of determining conformance with this specification, an observed or a calculated value shall be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of the Recommended Practices for Designating Significant Places in Specified Limiting Values (ASTM Designation: E29).

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Preparing activity:

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